Ji-Yeon Kim

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Research Interests

3D/4D Reconstruction and Generation of Humans, Objects, and Scenes
Monocular Reconstruction of Human and Object Motion; Collaborative Human–Object Interactions; Text-Conditioned 3D Avatar Generation; Depth-Aware Video Segmentation

Education

Pohang University of Science and Technology, POSTECH — Pohang, Korea
Integrated M.S.& Ph.D., Convergence of IT Engineering (Advisor: Prof. Tae-Hyun Oh) — Mar. 2020 – Present

University of Toronto — Toronto, Canada
Exchange Student (one semester), Faculty of Arts & Science — Jul. 2018 – Jan. 2019

Sookmyung Women's University — Seoul, Korea
Bachelor of Engineering, Major: IT Engineering — Mar. 2015 – Aug. 2019
Summa Cum Laude; Ranked 1st in the College of Engineering

Experience

International Computer Vision Summer School (ICVSS) — Sicily, Italy
Computer Vision for Spatial Intelligence — Jul. 2025

Awards & Honors

AFCV Best Robot Vision Paper Award, KRoC — Feb. 2024
"Depth-aware Video Panoptic Segmentation Model for Holistic Scene Understanding"

Minister's Award (1st Place, $12,000), Electronics Times ICT Paper Awards — Dec. 2023
"Text-driven 4D Human Avatar Generation via Cross-modal Synthesis-through-Optimization"

Qualcomm Innovation Fellowship Korea, Winner ($4,000) — Nov. 2022
"CLIP-Actor: Text-Driven Recommendation and Stylization for Animating Human Meshes"

Outstanding Poster Presentation Award, IPIU — Feb. 2022
"Generating Action-Conditioned 3D Human in Scenes"

National Scholarship for Science and Engineering — 2017 – 2019
Korea Student Aid Foundation

Academic Excellence Scholarship — 2015 – 2016
Sookmyung Women's University

Publications

(*An asterisk (*) denotes equal contribution.*)

Abbreviations

TPAMI	*IEEE Transactions on Pattern Analysis and Machine Intelligence*
RA-L	*IEEE Robotics and Automation Letters*
CVPR	*IEEE/CVF Conference on Computer Vision and Pattern Recognition*
ICCV	*IEEE/CVF International Conference on Computer Vision*
ECCV	*European Conference on Computer Vision*
ACCV	*Asian Conference on Computer Vision*
BMVC	*British Machine Vision Conference*
IROS	*IEEE/RSJ International Conference on Intelligent Robots and Systems*

Journal Articles

[J03] K. Youwang*, T. Byun*, **Kim Ji-Yeon**, S. Choi, T.-H. Oh, "CLIP-Actor-X: Text-driven 4D Human Avatar Generation via Cross-modal Synthesis-through-Optimization."
TPAMI 2026 (Minister's Award, ICT Paper Awards)

[J02] W. Jo, H. Ha, **Kim Ji-Yeon**, H. Jeong, T.-H. Oh, "LiDAR-Anchored Collaborative Distillation for Robust 2D Representations."
RA-L 2026

[J01] **Kim Ji-Yeon**, O. Hyun-Bin, K. Byung-Ki, D. Kim, Y. Kwon, T.-H. Oh, "Uni-DVPS: Unified Model for Depth-Aware Video Panoptic Segmentation."
RA-L 2024 **(Oral Presentation, IROS 2024)**

Conference Papers

[C10] A paper on 4D reconstruction of Multi-Human-Object interaction (Collaborative HOI)
Under review

[C09] W. Jo, H. Ha, **Kim Ji-Yeon**, H. Jeong, T.-H. Oh, "Self-Supervised Collaborative Distillation: Enhancing Lighting Robustness and 3D Awareness."
ICCV 2025 Workshop on Wild3D

[C08] W. Jo, K. Byung-Ki, **Kim Ji-Yeon**, H. Jeong, K. Joo, T.-H. Oh, "The Devil is in the Details: Simple Remedies for Image-to-LiDAR Representation Learning."
ACCV 2024

[C07] **Kim Ji-Yeon**, O. Hyun-Bin, K. Byung-Ki, D. Kim, Y. Kwon, T.-H. Oh, "Depth-aware Video Panoptic Segmentation Model for Holistic Scene Understanding."
Annual Conference of the Korean Robotics Society, 2024 (AFCV Best Robot Vision Paper Award)

[C06] L. Hyoseok, K. Jun-Seong, **Kim Ji-Yeon**, T.-H. Oh, "Feature Embedded 3D Gaussians for 3D Cognitive Map."
Workshop on Image Processing and Image Understanding, 2024

[C05] **Kim Ji-Yeon**, O. Hyun-Bin, D. Kim, T.-H. Oh, "MinDVPS: Minimal Model for Depth-aware Video Panoptic Segmentation."
CVPR 2023 Workshop on Vision Centric Autonomous Driving

[C04] K. Youwang*, **Kim Ji-Yeon***, T.-H. Oh, "CLIP-Actor: Text-Driven Recommendation and Stylization for Animating Human Meshes."
ECCV 2022 (Winner of the Qualcomm Innovation Fellowship Korea)

[C03] **Kim Ji-Yeon**, T.-H. Oh, “Generating Action-Conditioned 3D Human in Scenes.”
Workshop on Image Processing and Image Understanding, 2022 (Outstanding Poster Award)

[C02] K. Youwang, **Kim Ji-Yeon**, K. Joo, T.-H. Oh, “Unified 3D Mesh Recovery of Humans and Animals by Learning Animal Exercise.”
BMVC 2021

[C01] K. Youwang, **Kim Ji-Yeon**, K. Joo, T.-H. Oh, “Unified Pose and Shape Model for 3D Human-Animal Reconstruction.”
Summer Annual Conference of the Institute of Electronics and Information Engineers, 2021

Patent

US 20240273798A1
Text-driven motion recommendation and neural mesh stylization system and a method for producing human mesh animation using the same

KR 10-2886014
Method and apparatus for motion animating and mesh stylization using text-driven motion recommendation

KR 10-2459293
Method and apparatus for generating mesh model of human or quadrupeds

Academic Service

Journal Reviewer
IEEE Robotics and Automation Letters (RA-L), 2024, 2026
The Visual Computer (TVCJ), 2022

Conference Reviewer
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024, 2025
IEEE/CVF International Conference on Computer Vision (ICCV), 2025
International Conference on Learning Representations (ICLR), 2024, 2025, 2027
Conference on Neural Information Processing Systems (NeurIPS), 2024
IEEE International Conference on Robotics and Automation (ICRA), 2023